

Program	: Computer Engineering Program	Date	: 5 April 2026
Examiners	: Dr. Tarek salem Eng : Eman, Habiba,Basant	Duration	2 hours
Course/Code	: Database Management Systems / CSE112		

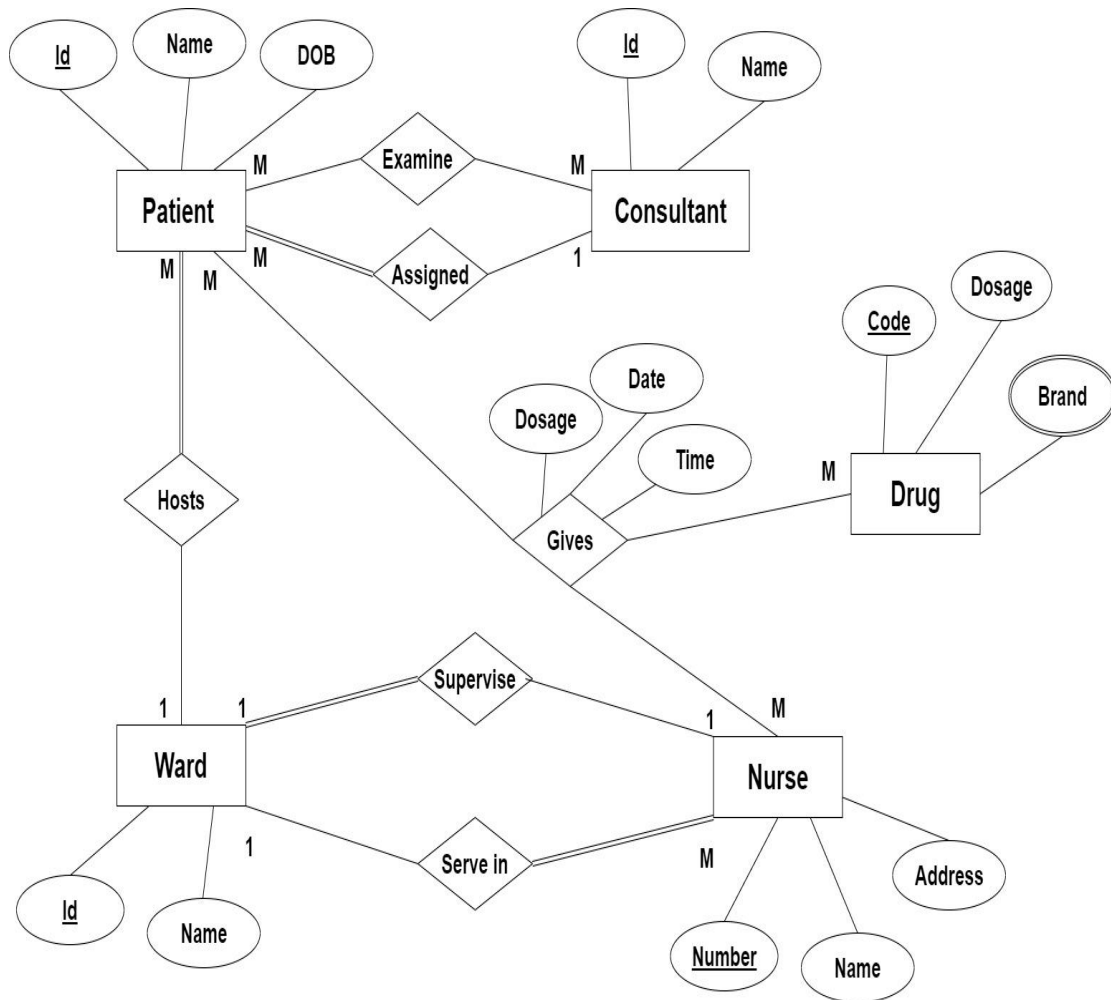
Question :

- A General Hospital consists of a number of specialized wards. Each ward is described by ward_id, Name
- The system records the following details about patients: Patient_id, name, Date_Of_Birth
- Each ward may host zero or more patients and each patient is hosted by only one ward.
- Each patient is assigned to one leading consultant but may be examined by other consultants, if required.
- Each consultant may be assigned zero or more patients and may examine zero or more patients.
- Consultants are described by Consultant_id, Name
- The system has to record all required data each time the Nurse gives a patient a certain drug with specified dosage at certain date and time.
- Each ward is under supervision of one nurse and a nurse may supervise only one ward.
- Each Nurse must serve in one ward
- Data about the nurse is recorded as her name and her number and her address.
- A drug has code number, recommended dosage and more than one brand name

Make ERD For This Hospital

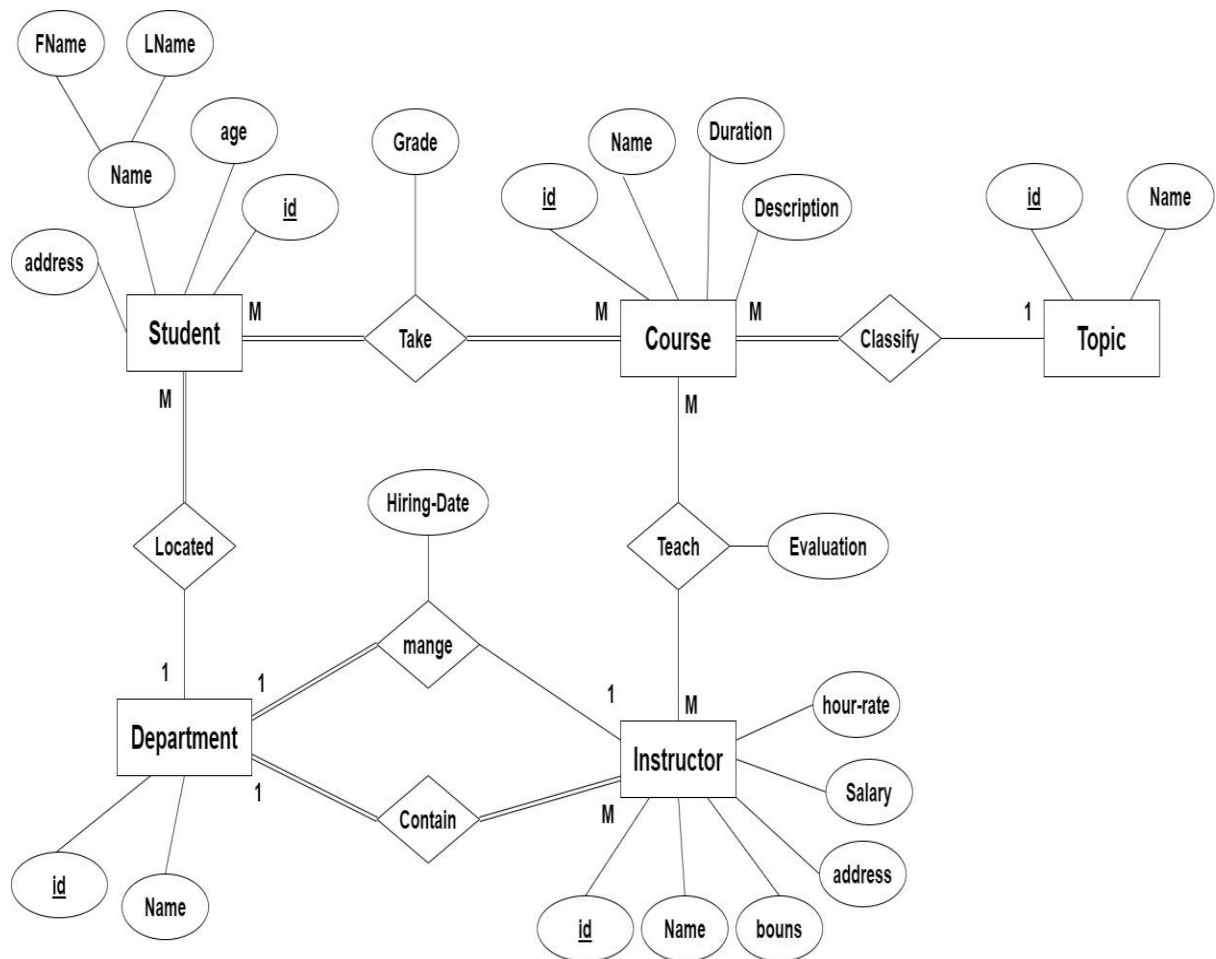
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ERD



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Create relational schema for the Following ERD ..



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Students

<u>Id</u>	FName	LName	Age	Address	<u>Dep_Id</u>
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Departments

<u>Id</u>	Name	Hiring_Date	<u>Ins_Id</u>
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Instructors

<u>Id</u>	Name	Address	Bouns	Salary	Hour_Rate	<u>Dep_Id</u>
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Courses

<u>Id</u>	Name	Duration	Description	<u>top_Id</u>
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Topics

<u>Id</u>	Name
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Stud_Course

<u>Stud_Id</u>	<u>Course_Id</u>	Grade
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Course_Instructor

<u>Course_Id</u>	<u>Inst_Id</u>	Evaluation
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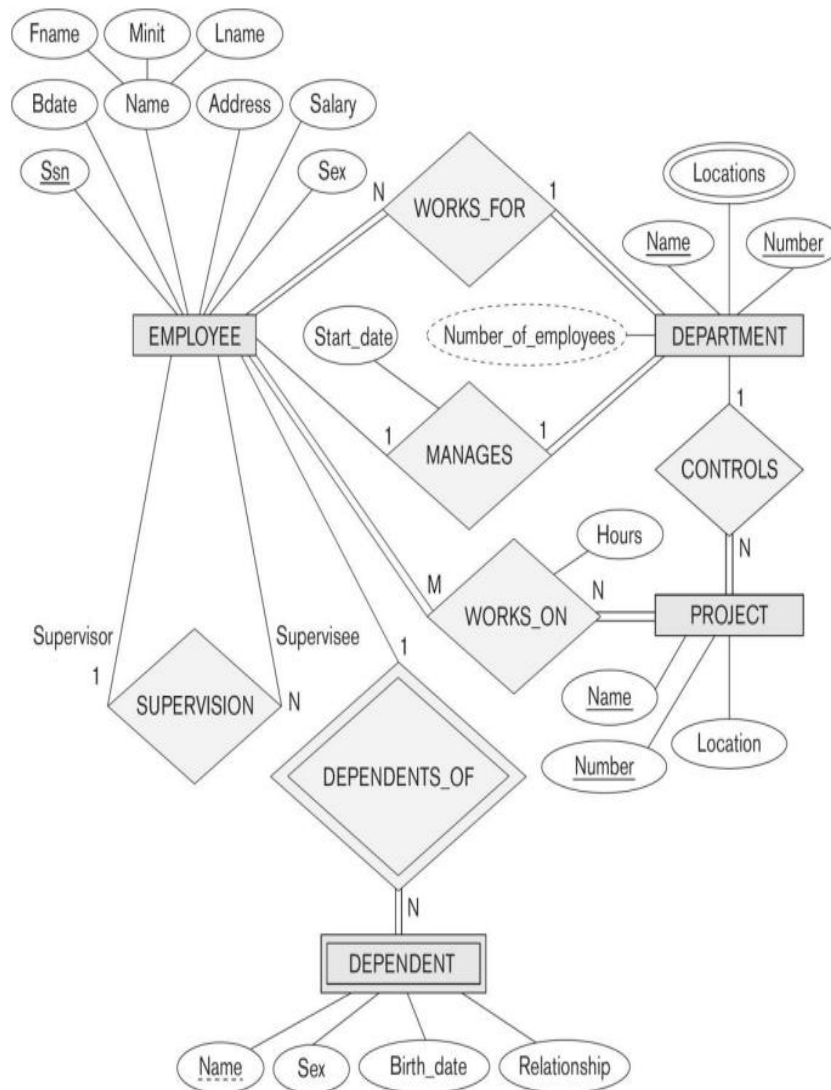
Question

A big company has decided to store information about its projects and employees in a database. The company has wisely chosen to hire you as a database designer. Prepare an E-R diagram for this Company according to The following Description:

- The company has a number of **employees** each employee has **SSN, Birth Date, Gender and Name** which represented as **Fname and Lname**.
- The company has a set of **departments** each department has a set of attributes **.DName, DNUM (unique) and locations**.
- Employees work in several **projects** each project has **Pname, PNumber as an identifier, Location and City**.
- Each employee may have a set of **dependent**; each dependent has **Dependent Name (unique), Gender, and Birth Date**.
Note: if the employee left the company no needs to store his dependents info
- For each Department, there is always one employee assigned **to manage** that Department and each manager has a **hiring Date**
- Department may have employees but employee must work on Only One department
- Each department may have a set of projects and each project must assigned to one department
- Employees work in several projects and each project has several employees and each employee has a number of working hours in each project
- Each employee has a supervisor

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Resulting Relational Schema

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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